

In vitro* methyl farnesoate secretion by mandibular organs isolated from different molt and reproductive stages of the crab *Oziotelphusa senex senex

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ABSTRACT: Considerable evidences indicate that mandibular organ (MO) of crustaceans secrete terpenoid hormone (methyl farnesoate [MF]), which regulates molting and reproduction. In the freshwater south Indian rice field crab *Oziotelphusa senex senex*, MO isolated during premolt and vitellogenesis, secreted greater amounts of MF, which indicates that the regulation of the crustacean molt and reproduction is complex and also involves MF, besides steroid (ecdysteroid) and peptide (sinus gland peptide) hormones.

KEY WORDS: crab, mandibular organ, methyl farnesoate, molting and reproduction, *Oziotelphusa*.

INTRODUCTION

Le Roux¹ was the first to identify the mandibular organ (MO) in crustaceans. Since then, the MO has been identified in a large number of crustaceans.^{2,3} Initial studies by several authors suggested that the MO might be involved in the regulation of reproduction^{1,4} and molting.^{5–7} This view was based on cyclical ultra-structural changes of the MO during molting and reproduction. The MO of decapod crustaceans synthesize and secrete methyl farnesoate (MF), a compound that is structurally similar to the juvenile hormone (JH) produced by the corpora allata of insects.^{8,9} The importance of JH in the regulation of metamorphosis, reproduction and behavior in insects^{10,11} has led to the hypothesis that MF may have similar functions in crustaceans. Though elucidation of the role of MF in crustacea is far from complete, recent studies indicate that MF may regulate reproduction,^{12–14} molting,^{15,16} metamorphosis^{8,17} and behavior.^{18,19}

High levels of MF in the hemolymph were correlated with vitellogenesis in females and reproductive morphogenesis in males.^{9,20–22} Stimulation of ovarian maturation in the crayfish *Procambarus clarkii* in response to dietary application of MF has been demonstrated.¹² The injection of MF also

stimulated ovarian growth in the freshwater south Indian rice field crab *Oziotelphusa senex senex* (*O. senex senex*)¹³ and in the prawn *Penaeus indicus*.²³ MF is also involved in the regulation of molting. Yudin *et al.*⁷ demonstrated a shortening of molt intervals of shrimp that had been implanted with MO. Tamone and Chang¹⁵ demonstrated that MF stimulates the secretion of ecdysone from Y-organs. Taken together, these observations indicate that MF functions to stimulate Y-organs and raises the question as to whether MF may also function as the endocrine signal that is responsible for molting. In the present study, the authors have measured the alteration in secretion of MF from the MO isolated from different stages of reproduction and molting of *O. senex senex*. The circulatory levels of MF during molting and reproduction in the crab were also measured.

MATERIALS AND METHODS

Animals

Freshwater south Indian rice field crabs *O. senex senex* were collected from paddy fields around Tirupati and maintained in large tanks at 22–23°C. Crabs were fed daily with sheep meat. In *O. senex senex*, molting and reproduction are antagonistic events. The entire reproductive cycle is completed during the intermolt period. In nature, premolt animals are always with quiescent ovaries.

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The molt and reproductive stages in *O. senex senex* was determined by the criteria described earlier^{24,25} for this crab. In *O. senex senex*, the previtellogenic (immature) ovary is translucent to opaque white. During vitellogenesis, the color of the ovary changes from pale yellow (vitellogenesis stage I) to orange (vitellogenesis stage II) and then it becomes dark brown (vitellogenesis stage III) prior to spawning. Molt stages were characterized based on setogenesis in the exopodite of third maxilliped.

Unlike eyestalk ablation experiments, it is not possible to do classical ablation-replacement studies to investigate the function of MO, since this gland is located at the base of the tendon associated with the posterior abductor muscle of the mandibles, and is difficult to surgically remove without killing the animal. To study the synthesis of MF by MO during molting and reproduction, the authors isolated MO from animals in different molt and reproduction stages and measured the synthesis of MF at different time points *in vitro*. Prior to isolation of MO, hemolymph was collected from these animals and used for MF determination.

Mandibular organ bioassay

Only uninjured female crabs (body weight 26–29 g) in different molt and reproductive stages were used in the present study. MO were dissected from ice-anesthetized crabs and washed in ice-cold modified medium 199 for 30 min. Medium 199 was prepared by dissolving 10.5 g of dried 199 medium (containing Earle's salts, L-glutamine without sodium bicarbonate) in 1 L of distilled water, and finally, sodium bicarbonate (2.2 g) was added to the medium after sterilization. The osmolarity of the medium used to culture the MO isolated from *O. senex senex* was standardized earlier.²⁶ MO were subsequently cultured in 24-well tissue culture plates in 2 mL of the same medium for up to 72 h at 18°C on shaker (50 r.p.m.) in a CO₂ incubator. Only one MO was cultured in one well. MF was extracted from the medium (1.0 mL) by adding 2.5 mL of CH₃CN, 0.5 mL of 4% NaCl (w/v), centrifuged at 1000 × g for 10 min at 4°C and partitioned against 1 mL of hexane. The dried hexane extract was assayed for MF using high performance liquid chromatography.²⁷ To determine circulatory MF levels, freshly drawn hemolymph samples (1.0 mL) were mixed with 2.5 mL of acetonitrile and 1.0 mL of 1% NaCl, and then extracted with hexane using the procedures described previously.¹⁵ In brief, samples containing MF were separated using μ Bondapak C₁₈ column (Waters, Milford, MA, USA) with an isocratic elution of 70% acetonitrile in

water (1.0 mL/min.). Peaks were detected at 254 nm by UV absorption. Retention times of the peaks were compared to those of standard MF. The MF was also confirmed by cochromatography with authentic all-*trans*-methyl farnesoate (gift from Prof. R. Ramamurthi, Senior Scientist, Department of Zoology, Sri Venkateswara University, Tirupati).

Data analysis

The data were analyzed statistically using the Dunnett's test to determine differences between control and experimental groups.

RESULTS

The secreted MF by MO isolated from immature intermolt (stage C4) at different time points is presented in Figure 1. Detectable levels of MF was observed after 3 h in culture and increased as time progressed. After 24 h, MF secretion from MO increases even further. The incubations were not carried out for more than 72 h. All subsequent incubations were only conducted up to 48 h.

The *in vitro* secretory rates of MF by MO were closely related to the stages of ovarian growth. MF secretion was lowest in the previtellogenic stage

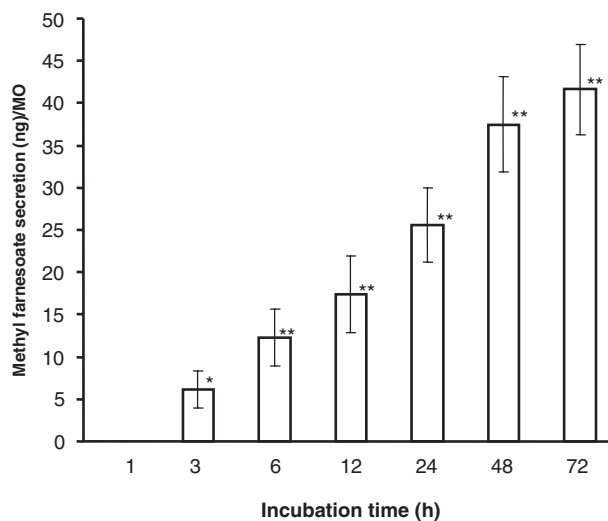


Fig. 1 Time course on the *in vitro* secretion of methyl farnesoate from mandibular organs (MO) isolated from previtellogenic, intermolt crabs. MO were dissected and incubated for 72 h in modified medium 199. Medium (50 μ L) was removed at each time point shown and assayed for methyl farnesoate. Values represent the mean $\pm$ standard deviation of five samples. Asterisks indicate significance at * P < 0.05; ** P < 0.001.

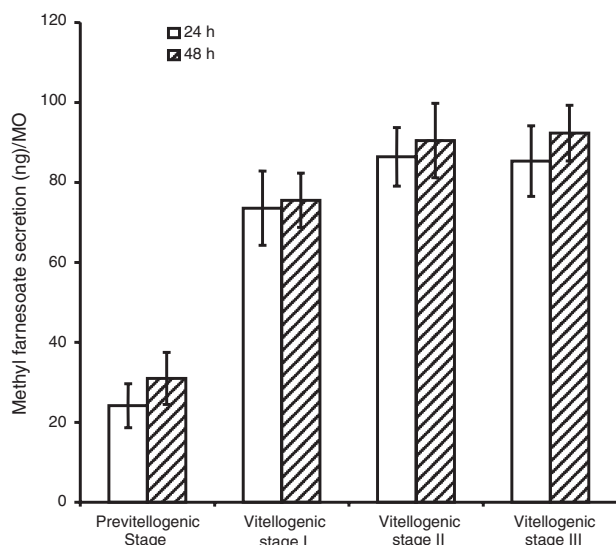


Fig. 2 Methyl farnesoate secretion by mandibular organs (MO) isolated during different reproductive stages of the crab *Oziotelphusa senex senex*. Values are the mean $\pm$ standard deviation of four samples. All values are significantly ($P < 0.0001$) different from previtellogenic stage.

and significantly increased as the vitellogenesis progressed, and maximum secretion was observed in the vitellogenesis stage III (Fig. 2). The results also indicate that more MF is secreted with increased incubation duration. MF secretion was lowest in MO isolated from postmolt (stage A) crab. The secretory rate was significantly high from MO isolated from premolt crabs (Fig. 3).

The MF levels in the hemolymph of *O. senex senex* during molting and reproduction is presented in Table 1. A gradual increase in the hemolymph MF concentration was observed as the crab approached premolt and reproduction.

DISCUSSION

Methyl farnesoate is nearly identical in structure to insect JH-III, differing only by the absence of an epoxide group. The functions of JH in insects were originally investigated by ablation and implantation of the glands that produce JH, the corpora allata. Removal of the corpora allata during the penultimate larval instar results in precocious metamorphosis, and implantation of corpora allata results in the formation of a supernumerary larval stage.²⁸ A major difficulty in investigating the roles of MF has been the anatomical location of the MO, which has made its surgical removal difficult. Therefore, it has not been possible to do 'classical' ablation-replacement studies to investigate MF

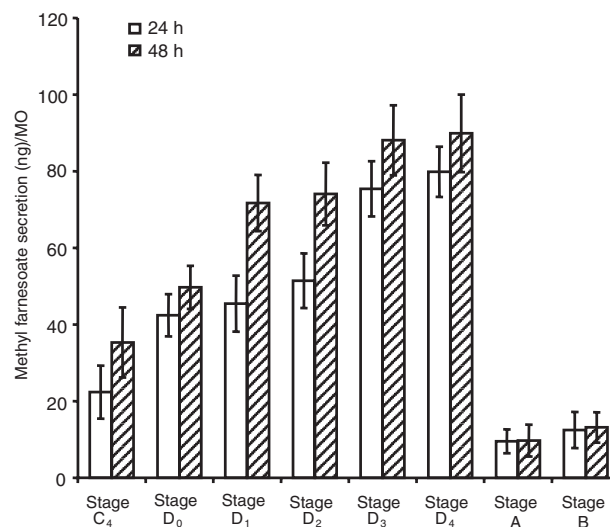


Fig. 3 Methyl farnesoate secretion by mandibular organs (MO) isolated during different molt stages of the female crab *Oziotelphusa senex senex*. Values are the mean $\pm$ standard deviation of four samples. All values, except stage B, are significantly ($P < 0.0001$) different from molt stage A.

Table 1 The levels of methyl farnesoate in hemolymph of *Oziotelphusa senex senex* during molting and reproduction

Stage	Methyl farnesoate level (ng/mL)
Molting	
A	—
B	11.4 $\pm$ 3.3 (–30.06)
C ₄	16.3 $\pm$ 3.1
D ₂	30.8 $\pm$ 2.9 (88.95)
D ₃	35.5 $\pm$ 4.2 (117.79)
D ₄	42.8 $\pm$ 4.6 (162.57)
F-value	64.017
P	<0.0001
Reproduction	
Immature ovaries	16.3 $\pm$ 3.1
Vitellogenesis I (translucent ovaries)	22.6 $\pm$ 4.6 (38.65)
Vitellogenesis II (pale yellow color ovaries)	44.1 $\pm$ 4.4 (170.55)
Vitellogenesis III (dark brown color ovaries)	56.7 $\pm$ 6.2 (247.85)
F-value	79.570
P	<0.0001

Values are mean ($n = 5$) $\pm$ standard deviation.

Values in the parentheses represent percentage change from C₄ or immature ovaries.

function. As a result, most investigators have sought to demonstrate a correlation between a physiological event and either the synthetic rate of MF by the MO or hemolymph levels of MF.

The secretory rate of MF by the MO isolated during premolt and vitellogenesis is significantly greater compared to secretory rate of MO isolated from intermolt and previtellogenesis stage crabs. Evidence for a role played by MF in crustacean reproduction and molting is accumulating. Implantation of MO led to premature vitellogenesis in the female juvenile spider crab *Libinia emarginata*.²⁹ In the same crab, MO synthesis and secretion of MF was synchronous with the vitellogenic cycle.⁹ Circulatory levels of MF in the crab *Cancer pagurus*,³⁰ crayfish *Procambarus clarkii*,¹² and in the Norway lobster *Nephrops norvegicus*³¹ showed an increase and correlated with the onset of secondary vitellogenesis. The addition of 1 ng MF/prawn per day in the diet enhanced the fecundity in the prawn *Penaeus vennamei*.³² Stimulation of ovarian growth in *Procambarus clarkii*,^{12,14} *O. senex senex*,^{13,33} and *Penaeus indicus*²³ was observed after administration of MF. Mark *et al.*³⁴ reported the induction of expression of vitellogenin in the hepatopancreas of female crabs *Charybdis feriatus* by farnesoic acid, a precursor for the MF.

Several lines of evidence also suggest that MF influences the molt cycle in certain crustaceans. A significant acceleration of the molt cycle accompanied by frequent molting occurred in the shrimp *Penaeus setiferus* that were implanted with MO isolated from the blue crab *Callinectes sapidus*.⁷ Similar results were observed following injection of *Procambarus clarkii* MO homogenates into the shrimp *Caridina denticulata*.³⁵ Wilder *et al.*³⁶ reported MF titers in the hemolymph peaks during premolt in *Macrobrachium rosenbergii*. Abdu *et al.*³⁷ observed that administration of MF into wintering *Cherax quadricarinatus* females accelerated molting.

The greater levels of MF secreted by MO, isolated during premolt and during vitellogenesis, indicates the role played by MF in regulating these processes. Besides, Tamone and Chang¹⁵ discovered that *in vitro* ecdysteroid secretion by the Y-organs of *Cancer magister* was stimulated by 1 μ M MF, and also by coculture with MO. These results strongly suggest that MF may function as an ecdysiotropin. The increased secretory levels of MF from the MO isolated from premolt and vitellogenic stage III crabs observed in the present study might stimulate the Y-organs to synthesize and secrete ecdysteroids. Although ecdysteroids are known as molting hormones, they also accumulate in the ovaries^{38–40} and stimulate oocyte maturation.⁴¹

In conclusion, it has been shown in this study that the MF synthesis is greater during premolt and vitellogenesis, which provides strong evidence that MF is indeed a growth hormone and also a reproductive hormone in Crustacea. Further experi-

ments, which are needed to firmly establish the role of MF in regulating the levels of ecdysteroids in *O. senex senex* are in progress in this laboratory.

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